Supporting Information

Efficient amino acid-based reactive capture of CO_2 via nickel molecular catalyst

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Supplementary Table 1| Diffusion-coefficients in m^2s^{-1}.

Constant	Value	References
D_{H^+}	9.311×10^{-9}	1
D_{K^+}	1.957×10^{-9}	
D_{OH^-}	5.273×10^{-9}	
D_{CO_2}	1.91×10^{-9}	
$D_{HCO_3^-}$	1.185×10^{-9}	
$D_{CO_3^{2-}}$	0.923×10^{-9}	

Supplementary Table 2| Solvation sizes in m.

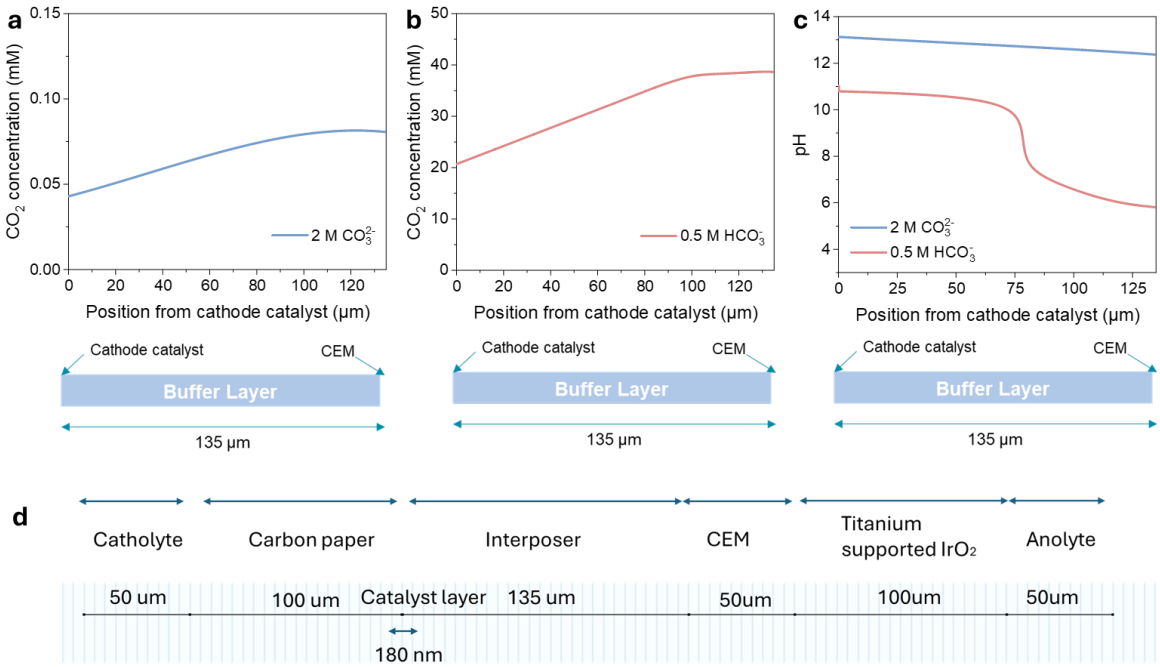
Constant	Value	References
a_{H^+}	0.56×10^{-9}	1
a_{K^+}	0.662×10^{-9}	
a_{OH^-}	0.6×10^{-9}	
a_{CO_2}	0.23×10^{-9}	
$a_{HCO_3^-}$	0.8×10^{-9}	
$a_{CO_3^{2-}}$	0.788×10^{-9}	

Supplementary Table 3| Rate constants used for modelling.

Constant	Value	Units	References
k_{1f}	3.71×10^{-2}	s^{-1}	
k_{1r}	1.99×10^{4}	s^{-1}	
k_{2f}	59.99	s^{-1}	
k_{2r}	2.28×10^{7}	$mol^{-1}m^{3}s^{-1}$	
k_{3f}	2.23	$mol^{-1}m^{3}s^{-1}$	1
k_{3r}	5.23×10^{5}	s^{-1}	
k_{4f}	6.0×10^{6}	$mol^{-1}m^{3}s^{-1}$	
k_{4r}	1.07×10^{6}	s^{-1}	
k_{wf}	2.4×10^{-2}	$mol\ m^{3}s^{-1}$	
k_{wr}	2.4×10^{6}	$mol^{-1}m^{3}s^{-1}$	
k_{5f}	2.29×10^{-1}	$mol\ m^{-3}s^{-1}$	2

Supplementary Table 4| h parameters used to estimate Sechenov's constant in $m^{3}kmol^{-1}$.

Constant	Value	References
h_{K_+}	0.0922	
h_{OH_-}	0.0839	
h_{HCO3^-}	0.0967	3
$h_{CO3^{2-}}$	0.1423	
$h^{0}_{CO_2}$	-0.0172	
$h^{T}_{CO_2}$	-0.00038	



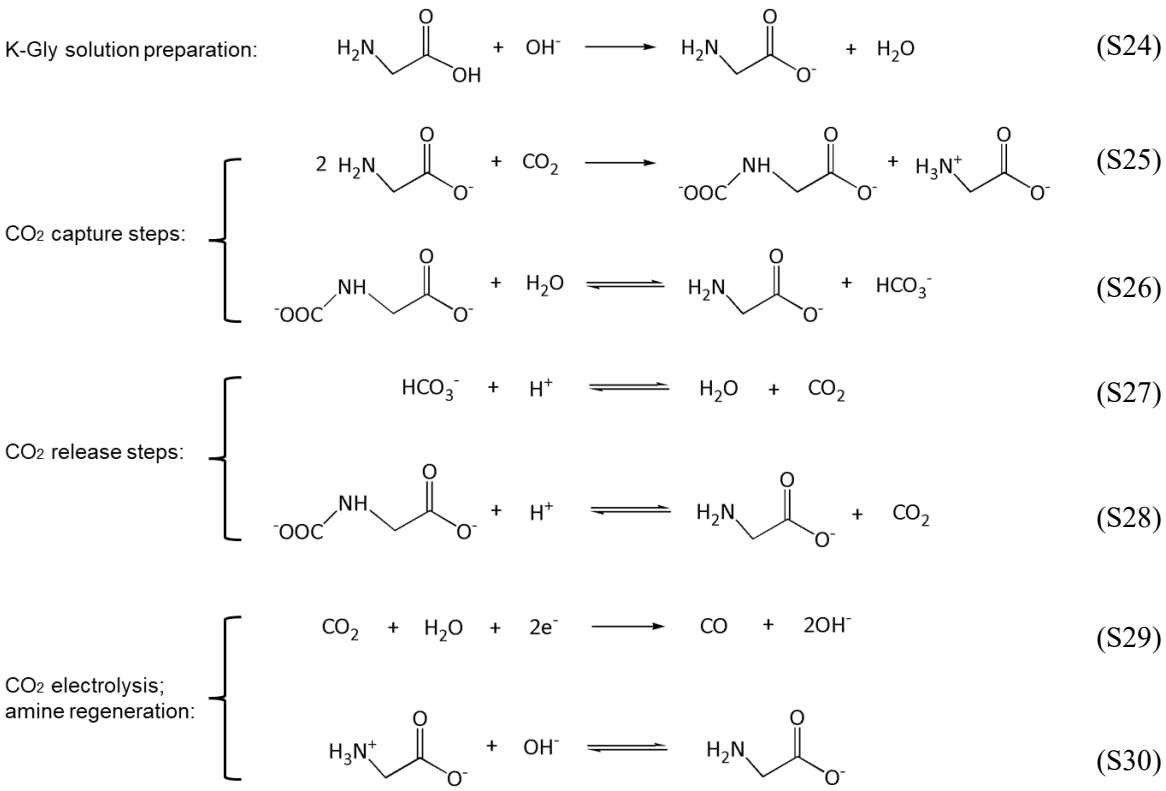
Supplementary Figure 1 | COMSOL simulations of CO_2 concentration in (a) 2 M carbonate electrolysis, (b) 0.5 M bicarbonate electrolysis. (c) COMSOL simulation of pH in 2 M carbonate and 0.5 M bicarbonate solutions. (d) Multiphysics simulation of carbonate/bicarbonate electrolysis in one dimension.

Supplementary Table 5 | Capture solution comparison (the concentration of capture solution was fixed at 1.5 M for all tests). Cost was obtained from Sigma for purities ⩾ 98% and at the lowest unit price.

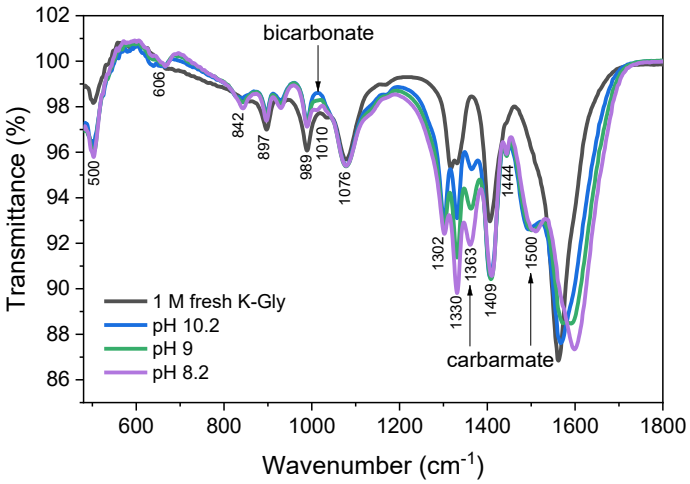
	Amino acid salt (K-Gly)	**NaOH**	**Na_2CO_3**	**Amine (monoethanolamine)**
O_2 tolerance (mmol mol^{-1})[4]	98.33	100	100	97.08
CO_2/e	1	0.5	1	1
CO_2 loading (mmol mol^{-1}) in 24h[5]	89.6	88.1	16.9	87.3
Capture rate (mmol mol^{-1} h^{-1})[5]	3.73	3.67	0.7	3.63
Toxicity (Lethal dose) (mg kg^{-1})[6]	7930	340	2800	1089
Cost (\$ kg^{-1})	43	59	67	14

Supplementary Note 1 | Chemical reactions of CO_2 capture and K-Gly regeneration in the reactive capture.

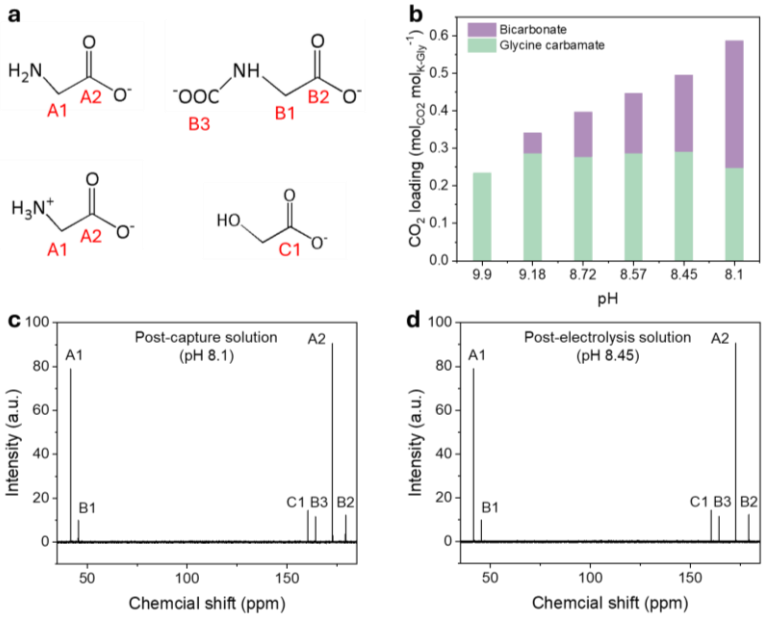
Chemical reactions of CO_2 capture and K-Gly regeneration in K-Gly-based reactive capture were shown in Supplementary Figure 2. K-Gly was prepared by mixing KOH and glycine to generate glycinate at a 1:1 molar ratio (see equation 24). Glycinate then reacts with CO_2 to produce carbamate and glycine zwitterion (see equation S25). The carbamate subsequently undergoes hydrolysis to yield glycinate and bicarbonate at a sufficiently high CO_2 loading (see equation S26). The hydroxyl groups produced locally on the cathodic catalyst react with glycine zwitterion to regenerate glycine (see equation S30). During reactive capture, protons are generated by the anodic oxygen evolution and transported to cathode via a cation exchange membrane (Figure 1b). The protons then react with bicarbonate ions to produce CO_2 following with passing through a buffer layer[7]. In situ generated CO_2 is subsequently electrochemically reduced on the cathode.



Supplementary Figure 2 | Chemical reactions of CO_2 capture and K-Gly regeneration in the reactive capture using K-Gly as the capture solution.



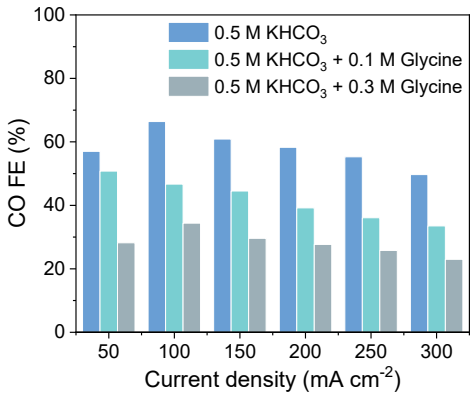
Supplementary Figure 3 | FTIR spectra of glycine-based capture solutions at different pH values after CO_2 purging. The fresh capture solution is prepared by mixing 1 M KOH, 1 M glycine, and 0.1 M KH_2PO_4. The peaks at 1076 cm^{-1} relates to KH_2PO_4[8]. The peaks at ~1500 cm^{-1} and ~1363 cm^{-1} correspond to the stretch of carbamate, while the peaks at ~1409 cm^{-1} and ~1600 cm^{-1} correspond to the stretch of carboxylate[9]. The peaks at ~1444 cm^{-1} and 1010 cm^{-1} relate to carbonate and bicarbonate respectively[10]. We found that the carbonate does not exist in the fresh capture fluid and exists at the higher pH values, while the bicarbonate peaks can only be observed for the capture solutions that captured CO_2 for a longer time when the pH value is lower than 9.0.



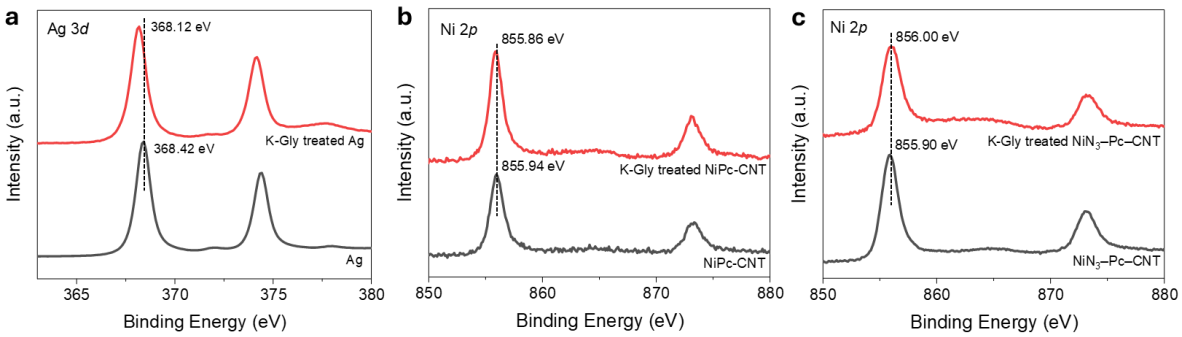
Supplementary Figure 4 | (a) Chemical structures, (b) concentrations of each component in fresh 1 M K-Gly solution, post-capture solution (different pHs) and the solution after 10-h stability (pH 8.45), ^{13}C NMR spectra of (c) post-capture solution and (d) post-electrolysis solution.

Supplementary Table 6 | Compositions in post-capture 1 M K-Gly solutions with different pHs (calculated from NMR results).

pH	Bicarbonate	Glycinate	Glycine carbamate	Glycine zwitterion
9.9	0	0.53182	0.23409	0.23409
9.18	0.05354	0.3717	0.28738	0.34092
8.72	0.11896	0.32502	0.27801	0.39697
8.57	0.15857	0.26593	0.28775	0.44632
8.45	0.2041	0.2126	0.29165	0.49575
8.1	0.33853	0.16403	0.24872	0.58725



Supplementary Figure 5 | Effect of glycine on the performance of Ag catalyst in reactive capture to produce CO at different current densities, using different capture solutions (first using 0.5 M $KHCO_3$; then adding additional 0.1 M glycine; and adding another 0.2 M glycine (0.3 M glycine in total) with 0.1 M KH_2PO_4 as the supporting electrolyte.



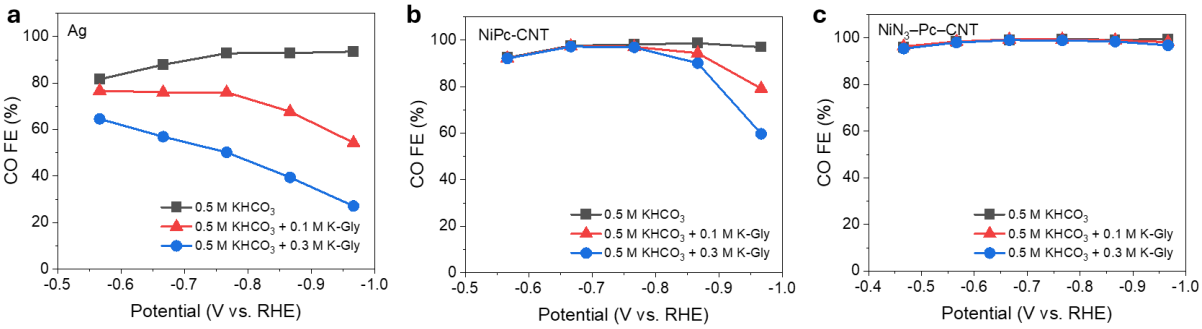
Supplementary Figure 6 | (a) high-resolution XPS Ag 3*d* of fresh and K-Gly treated Ag, (b) high-resolution XPS Ni 2*p* of fresh and K-Gly treated NiPc–CNT; and (c) high-resolution XPS Ni 2*p* of fresh and K-Gly treated NiN_3–Pc–CNT.

Supplementary Note 2 | CO_2RR in H-cell using different catholytes.

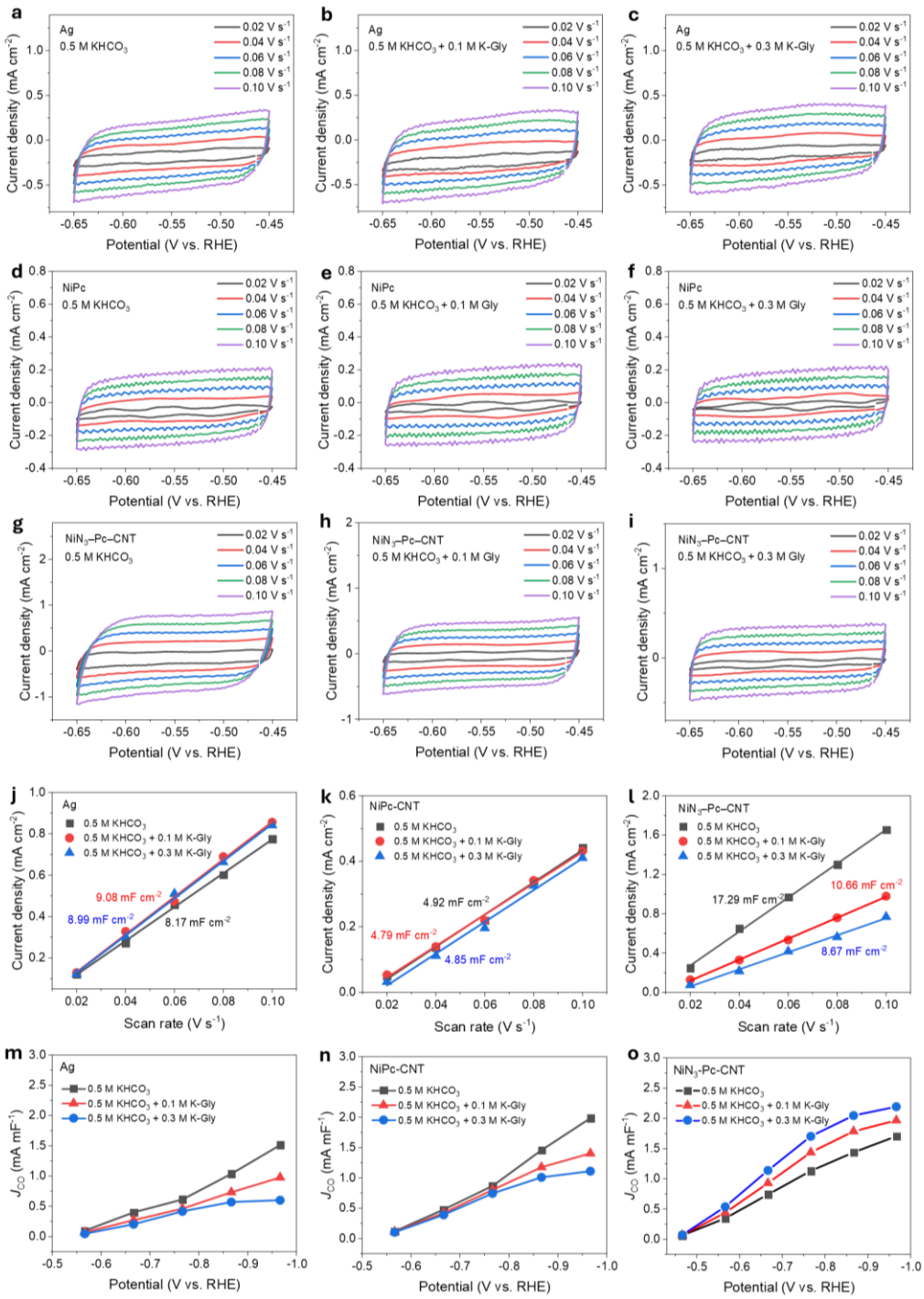
We also conducted in H-cell using 0.5 M $KHCO_3$ with different concentrations of K-Gly, allowing us to investigate the effect of glycine on the catalytic performance (Supplementary Figure 7). We also explored the impact of glycine on the intrinsic activity of the catalyst by calculating their electrochemical active surface areas (ECSA) via scanning cyclic voltammetry in the non-faradic region and normalizing the CO partial current density against the ECSA (Supplementary Figure 8).

Both CO FE and ECSA-normalized CO partial current density of Ag revealed deteriorations when K-Gly was added (Supplementary Figure 7a and Supplementary Figure 8m). We assume that the catalyst with strong amino acid adsorption such as Ag could suffer from activity deterioration under glycine input. However, the low local CO_2 concentration in H-cell could also impact the ECSA-normalized CO partial current density, explaining why the amino-acid-free NiPc–CNT also exhibited some degradations especially at high current densities due to its weak *COOH adsorption (Supplementary Figure 7b and Supplementary Figure 8n).

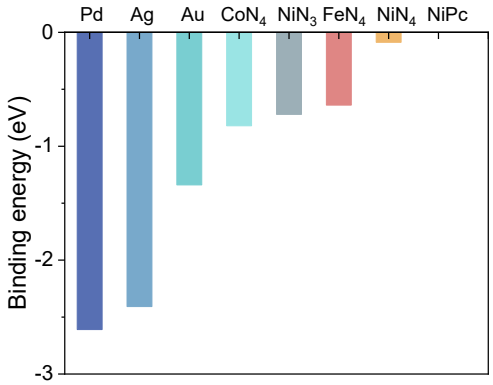
The issues (amino acid adsorption and limited local CO_2 concentration) did not exist for our catalyst, NiN_3–Pc–CNT (Supplementary Figure 7c and Supplementary Figure 8o). We concluded the remained FE upon the addition of K-Gly is due to its amino-acid-free property and strong *COOH adsorption for CO_2RR. Interesting, NiN_3–Pc–CNT exhibited increased ECSA-normalized CO partial current density when K-Gly was added. The added K-Gly might function as the capture promoter which possibly enrich the local CO_2 concentration to contribute its increased ECSA-normalized CO partial current density.



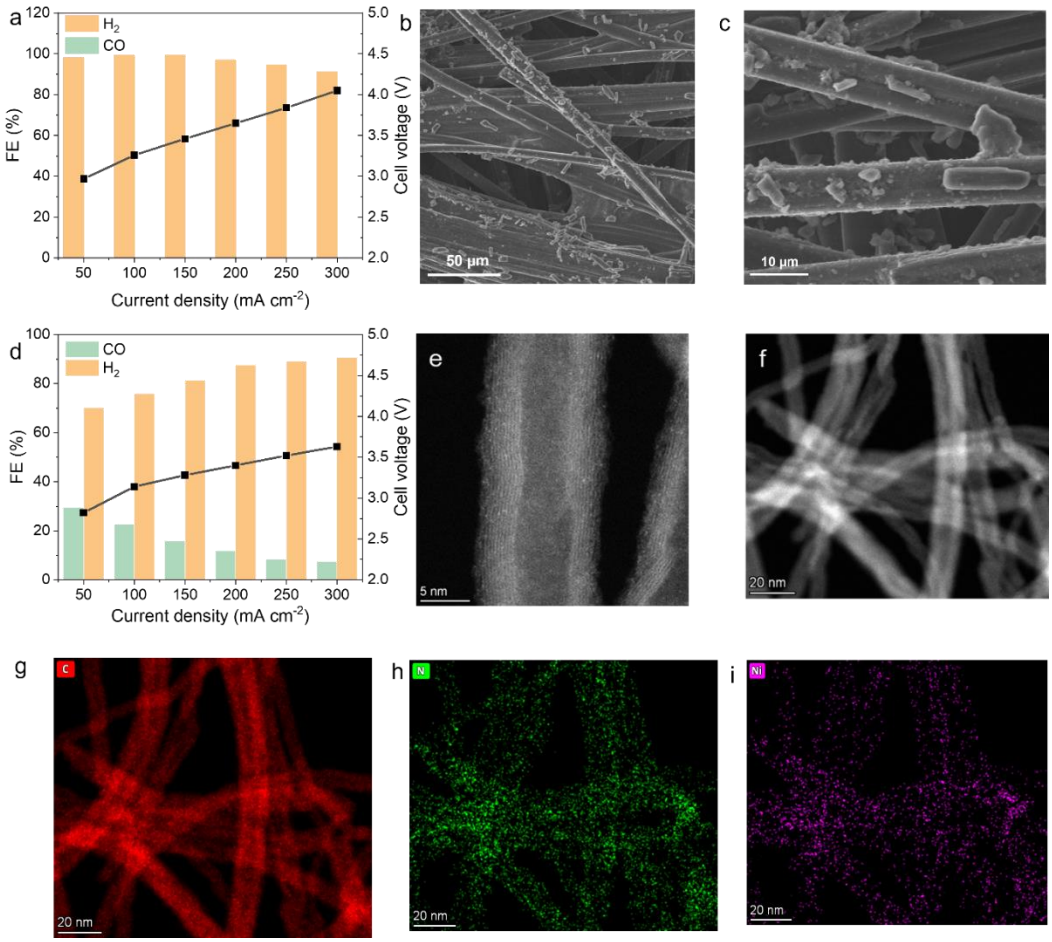
Supplementary Figure 7 | CO_2RR performance comparison of (a) Ag, (b) NiPc–CNT and (c) NiN_3–Pc–CNT at various potentials using different cathode electrolysers (0.5 M $KHCO_3$; 0.5 M $KHCO_3$ and 0.1 M K-Gly; 0.5 M $KHCO_3$ and 0.3 M K-Gly). Voltages are not iR-corrected.



Supplementary Figure 8 | Cyclic voltammograms with different scan rates of (a–c) Ag, (d–f) NiPc–CNT, and (g–i) NiN_3–Pc–CNT using 0.5 M $KHCO_3$, 0.5 M $KHCO_3$ and 0.1 M K-Gly, and 0.5 M $KHCO_3$ and 0.3 M K-Gly as the cathode electrolyser, respectively. Double-layer capacitance of (j) Ag, (k) NiPc–CNT, and (l) NiN_3–Pc–CNT; ECSA-normalized CO current density under different potentials of (m) Ag, (n) NiPc–CNT, and (o) NiN_3–Pc–CNT. All measurements in H-cell are performed in CO_2-saturated solution, and the catalyst mass loading is 0.5 mg cm^{-2}. Voltages are not iR-corrected.



Supplementary Figure 9 | DFT-computed binding energy of glycine molecule on conventional CO-producing catalysts. The binding energy between NiPc and glycine was considered as zero for reference.



Supplementary Figure 10 | (a) Reactive capture performance of NiPc and (b,c) SEM images in different magnifications. (d) Reactive capture performance of NiPc–CNT, (e,f) Scanning Transmission electron microscopy (STEM) images of NiPc–CNT, and (g–i) the corresponding EDS images.

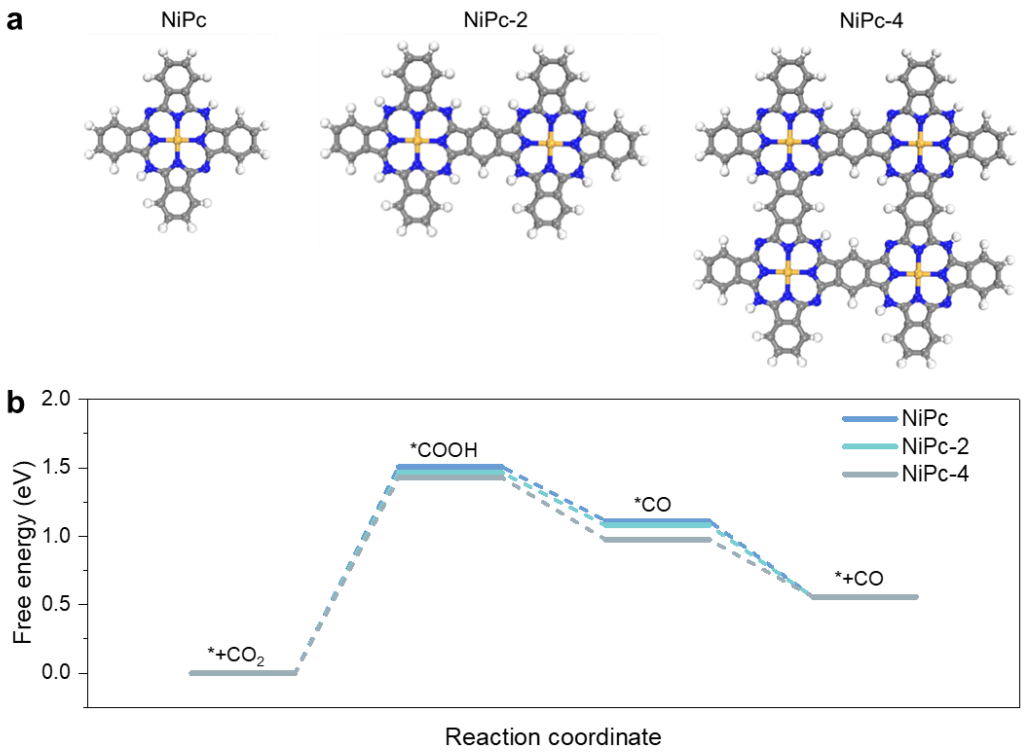
Supplementary Note 3 | DFT calculations for NiPc–CNT and NiN_3–Pc–CNT.

DFT results indicated that the first proton/electron transfer step, i.e., $*CO_2 + e^- + H^+ \rightarrow *COOH$, is the potential-limiting step for NiPc catalyst. This conclusion is supported by the highest Gibbs free energy increase observed during the $*CO_2$-to-*COOH step, suggesting that CO_2 activation (formation of *COOH) limits the catalytic activity of NiPc catalysts (Supplementary Figure 11). Since NiN_3–Pc–CNT has conjugated NiPc network and unsaturated Ni–N_3 structure, both of which could benefit for its performance improvement.

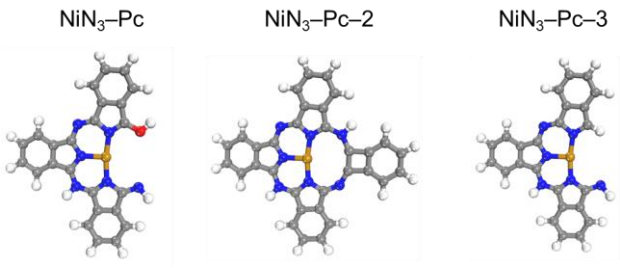
To investigate the impact of the conjugated structure, we studied three NiPc structures (depending on the number of NiPc units), named NiPc, NiPc–2, and NiPc–4 (Supplementary Figure 11a). Our results indicated that the formation of *COOH (the potential-determining step) on the catalyst surface becomes slightly more favourable with increasing the number of NiPc units, with NiPc–4 showing the lowest Gibbs free energy of *COOH (Supplementary Figure 11b). This indicates that the conjugated network could potentially enhance CO_2-to-CO performance, albeit to a limited degree.

To explore the impact of Ni–N_3 structure, considering different configuration possibilities, we constructed three Ni–N_3 geometries, denoted as NiN_3–Pc, NiN_3–Pc–2 and NiN_3–Pc–3 (Supplementary Figure 12). All Ni–N_3 structures exhibited stronger Gibbs adsorption energy of *COOH than NiPc (Supplementary Figure 13) and consequently a superior CO_2-to-CO production activity. The promoted *COOH adsorption is associated with the negatively charged Ni–atom center on Ni–N_3 structures, suggested by the Bader charge calculations (Supplementary Figure 14). We synthesized NiN_3–Pc via nucleophilic attack that opens up the C≡N bond of 1,2,4,5-tetracyanobenzene (TCNB) while additional TCNB molecules were incorporated into the intermediate to form the unsaturated phthalocyanine structure[11]. We utilized ethanol/water to serve as the nucleophile, therefore possibly adding oxygen atoms from ethanol/water into the NiN_3–Pc.

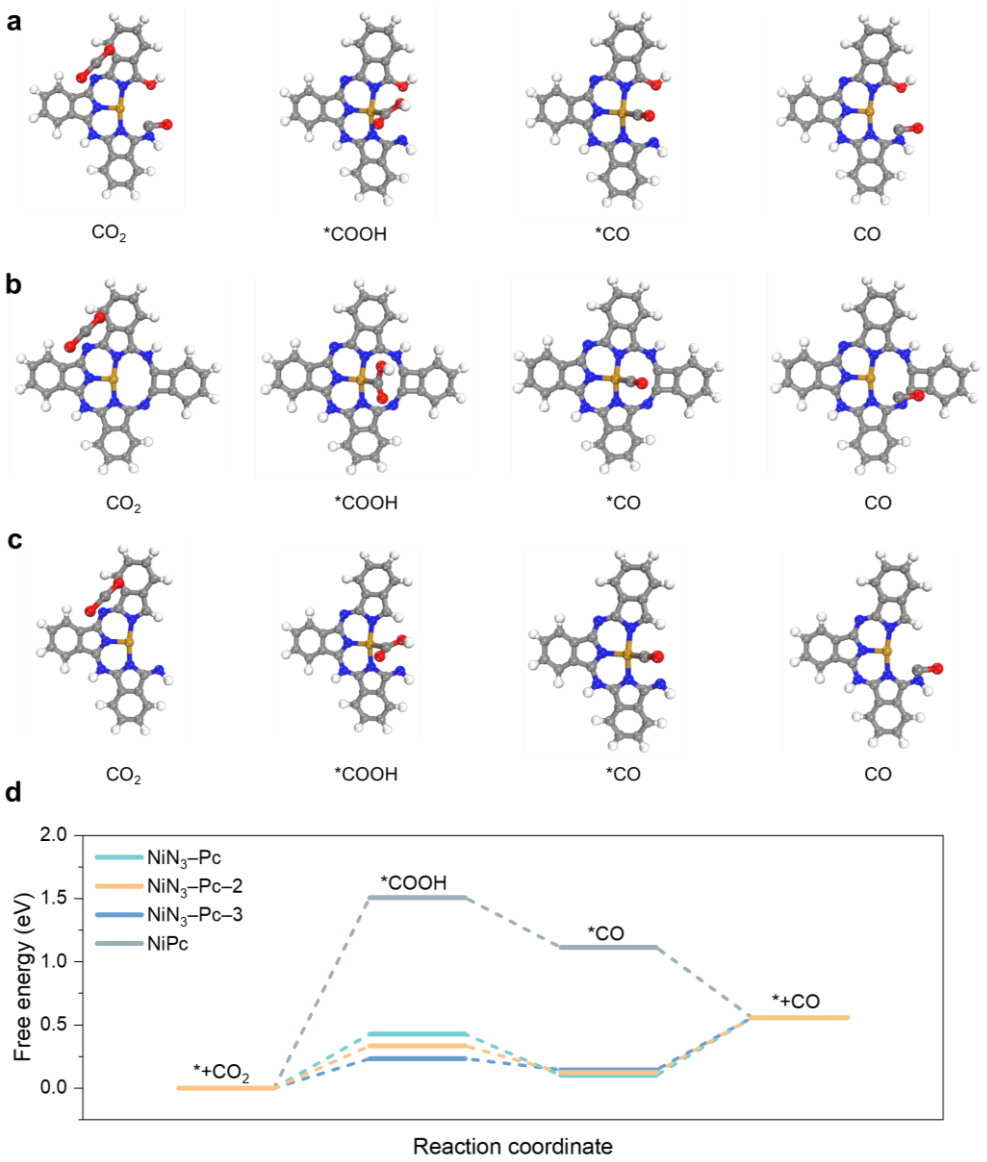
These DFT studies suggested that the enhanced performance of NiN_3–Pc–CNT is primarily attributed to the presence of conjugated NiPc network and, more importantly, coordinatively unsaturated Ni–N_3 structure.



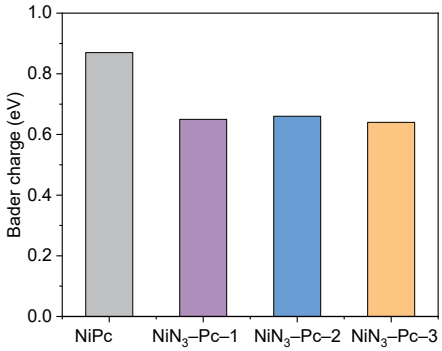
Supplementary Figure 11 | (a) Optimized structure of NiPc, NiPc–2 and NiPc–4. (b) Gibbs free energy diagram of electroreduction of CO_2 to CO on NiPc, NiPc–2, and NiPc–4.



Supplementary Figure 12 | Optimized structure of three possible NiN_3-Pc moieties (NiN_3–Pc, NiN_3–Pc–2 and NiN_3–Pc–3).



Supplementary Figure 13 | The reaction path of electrochemical reduction of CO_2 to CO on (a) NiN_3–Pc, (b) NiN_3–Pc–2, (c) NiN_3–Pc–3, and (d) the Gibbs free energy diagram of electrochemical reduction of CO_2 to CO on NiPc, NiN_3–Pc, NiN_3–Pc–2, and NiN_3–Pc–3.



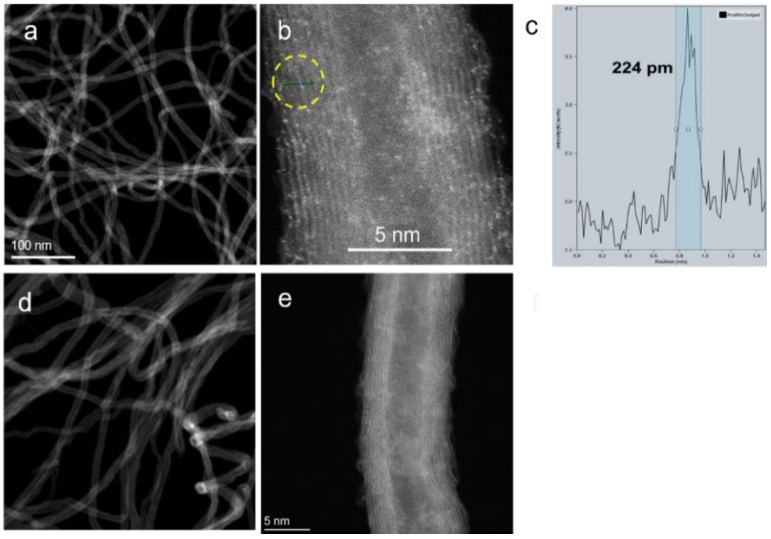
Supplementary Figure 14 | Bader charges of Ni atom in NiPc, NiN_3–Pc, NiN_3–Pc–2 and NiN_3–Pc–3.

Supplementary Table 7 | Contributions to Gibbs free energies of gas species and adsorbates. (zero-point energy (ZPE), enthalpic temperature correction ($\int C_P \mathrm{d}T$), entropy (S). H_2 is referring to the energy used in computational hydrogen electrode model).

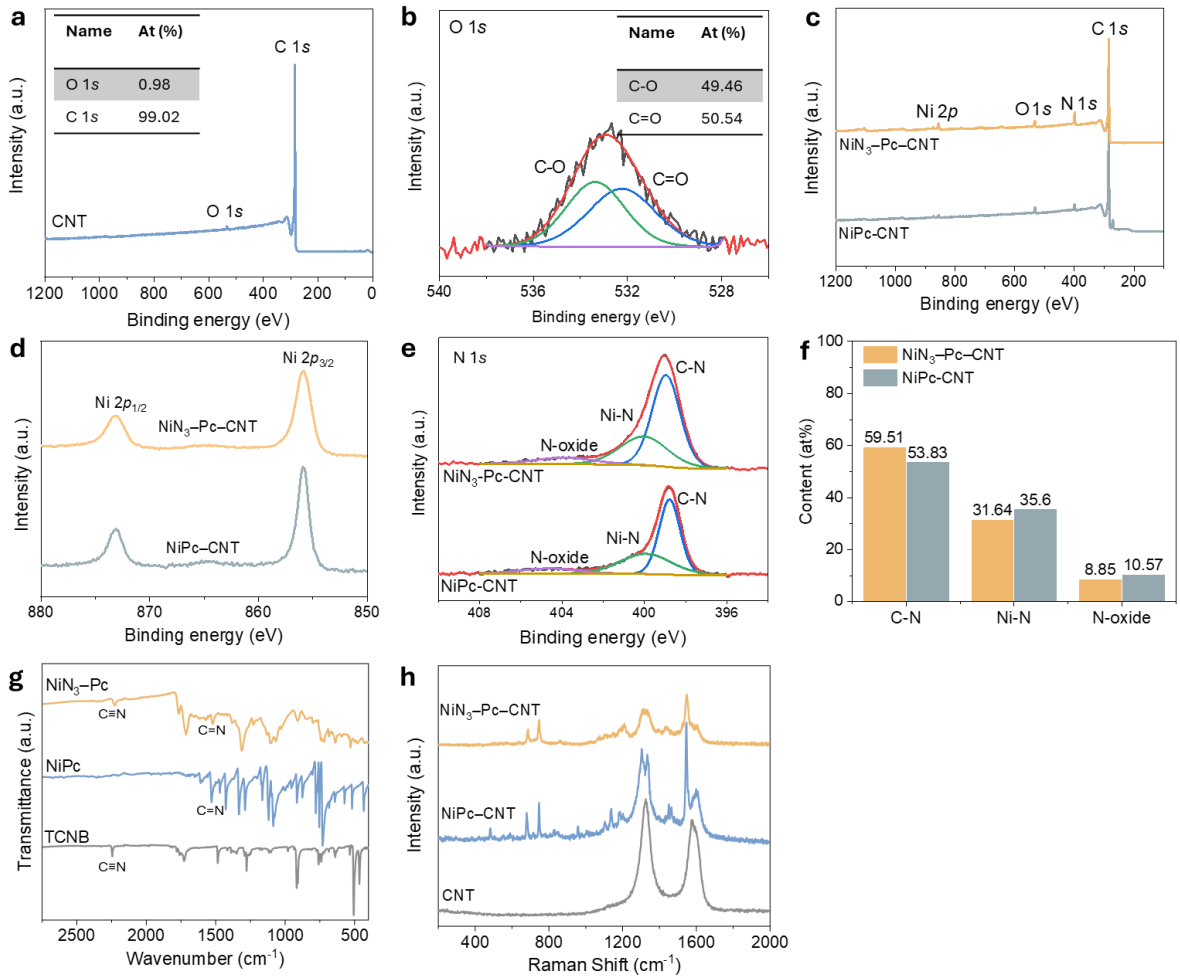
Species	$\boldsymbol{ZPE}$ $(\boldsymbol{eV})$	$\int \boldsymbol{C_P}\mathbf{d}\boldsymbol{T}$ $(\boldsymbol{eV})$	$-\boldsymbol{TS}$ (eV)	$\boldsymbol{G}-\boldsymbol{E}$ $(\boldsymbol{eV})$
CO_2(g)	0.31	0.10	-0.66	-0.25
H_2O(l)	0.57	0.10	-0.67	0
H_2(g)	0.27	0.08	-0.40	-0.05
$COOH^*$	0.61	0.09	-0.18	0.13
CO^*	0.18	0.06	-0.11	0.52

Supplementary Note 4 | Structure characterization of CNT and NiN_3–Pc–CNT

CNT and NiN_3–Pc–CNT were characterized by high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) to examine the morphology and the nickel dispersion. The samples were prepared by dispersing each powder ultrasonically in ethanol and subsequently drop-casting on Lacey Carbon 200 mesh copper TEM grids (Ted Pella, Inc.). All images indicate the presence of multi-walled CNTs, with an average diameter of around 12 nm. Occasionally, some coiled tubes can be found among the straight or bent-shaped majority. Compared with CNT (Supplementary Figures 15d–e), additional bright spots in Supplementary Figure 15b demonstrated successful deposition of atomically dispersed Ni sites on the surface of CNT. Supplementary Figure 15c shows the plot of intensity vs. position of the line profile indicating a width of around 224 pm at half intensity for the marked bright spot in Supplementary Figure 15b, minimizing the possibility of the presence of more than a single atom of Ni in NiN_3–Pc–CNT.



Supplementary Figure 15 | (a,b) HAADF-STEM images of NiN_3–Pc–CNT, and (c) the plot of intensity vs. position of the line profile. (d,e) HAADF-STEM images of CNT.

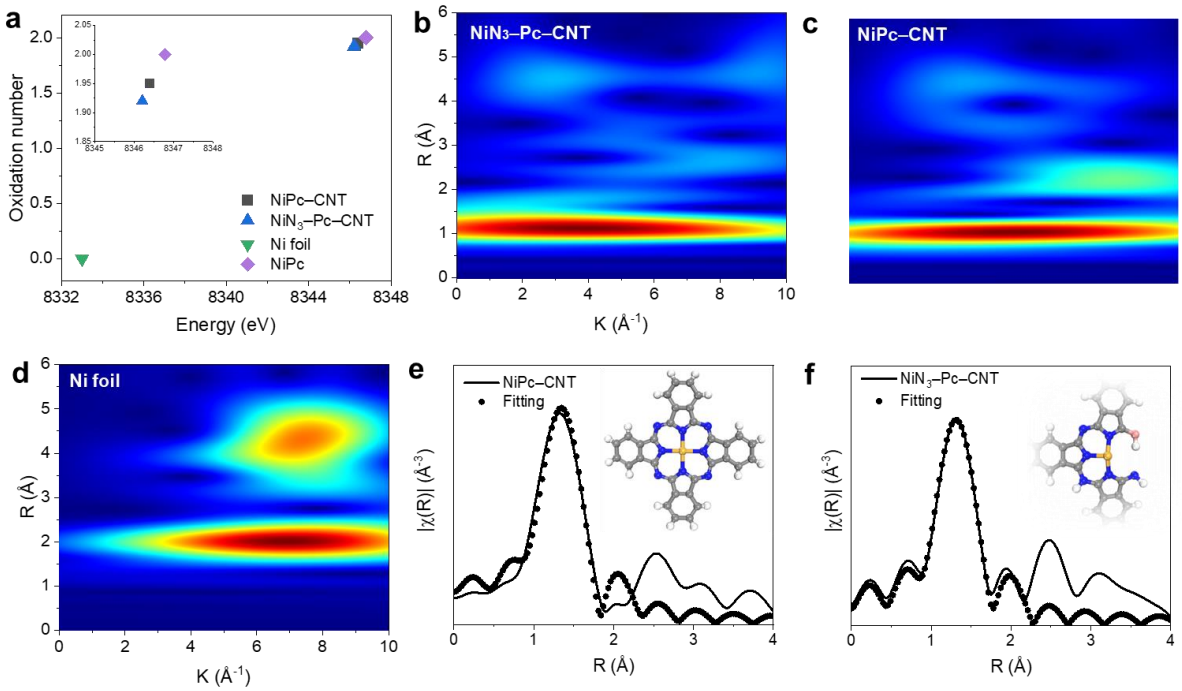


Supplementary Figure 16 | (a) XPS survey overall-spectrum of CNT and (inset) element composition, (b) O 1*s* region of CNT and (inset) oxygen contents. (c) XPS survey spectra and (d) high-resolution XPS Ni 2*p* of NiPc–CNT and NiN_3–Pc–CNT, (e) N 1*s* region and (f) nitrogen species determined by N 1*s* spectra of NiPc–CNT and NiN_3–Pc–CNT. O 1*s* XPS spectrum shows that CNT initially contains ~1 at% of O species, including C-O (C-OH and C-O-C) and C=O (COOH and C=O). NiPc–CNT has higher loadings of Ni-N bonds than NiN_3–Pc–CNT. A possible explanation is that nickel atom in NiN_3–Pc–CNT is surrounded by less nitrogen to get a smaller amount of Ni-N at the atomic ratio in these XPS results, which is in agreement with the XPS Ni 2*p* results (NiN_3–Pc–CNT has slightly lower electron density of nickel sites than NiPc–CNT in Supplementary Figure 16d). (g) FTIR spectra of TCNB, NiPc and NiN_3–Pc. (h) Raman spectra of CNT, NiPc–CNT and NiN_3–Pc–CNT. Relative to TCNB, NiN_3–Pc showed an attenuation of the C≡N band at 2225 cm^{-1} and a new C=N peak at 1520 cm^{-1}, corresponding to phthalocyanine (Supplementary Figure 16g)[12,13]. We observed skeletal vibrations between 400 and 1600 cm^{-1}, with strong bands at 1105 and 1312 cm^{-1}, attributed to pyrrole ring bending and C–C stretching vibrations, respectively.

Supplementary Note 5 | The XAS analysis of NiN_3–Pc–CNT and NiPc–CNT

Synchrotron-radiation-based X-ray absorption fine structure (XAS) analysis was conducted to investigate the electronic structure and coordination environment. Figure 3f shows the X-ray absorption near-edge structure (XANES) spectra of NiN_3–Pc–CNT and NiPc–CNT references. The absorption edge of Ni in NiN_3–Pc–CNT displays lower intensity than Ni foil but slightly higher than NiPc–CNT, showing the unique electronic structure of NiN_3–Pc–CNT is close to Ni^{2+}. To more specific, we did the XANES-linear fittings where all catalysts indicate quite similar oxidation state (close to +2), while NiN_3–Pc–CNT performed a slightly lower oxidation state than NiPc–CNT (Supplementary Figure 17a), in line with XPS results (Supplementary Figure 16d). The enhanced electronic localization at the Ni active site has been shown to improve catalytic performance by increasing CO_2 adsorption and activation[14,15]. Higher electronic localization at the Ni site in NiN_3–Pc–CNT can promote CO_2 adsorption and activation to enhance the CO_2RR activity. The white-line peak in the Ni K-edge spectrum of NiN_3–Pc–CNT shifts to higher intensity compared to NiPc–CNT, which also demonstrated the more orbital hybridization of the Ni centre in NiN_3–Pc–CNT.

The Ni K-edge of prepared catalysts was conducted to examine the Fourier transformed (FT) extended X-ray absorption fine structure (FT-EXAFS) spectra (Figure 3g). The Ni spectrum of NiN_3–Pc–CNT in EXAFS analysis shows Ni-N path and reveal no metallic nickel, indicating that the nickel molecular structure exists on the surface of CNT. The wavelet transform (WT) contour plots of NiN_3–Pc–CNT/NiPc–CNT display one intensity maximum at about 4 $Å^{-1}$ in K space, by comparing with the WT analysis of Ni foil with much higher intensity maximum (Supplementary Figure 17b-d). The intensity maximum at about 7 $Å^{-1}$ related to Ni-Ni coordination in Ni foil is not detected in NiN_3–Pc–CNT. The quantitative least squares EXAFS curve-fitting analysis was performed to extract the coordination configuration, finding that each Ni atom in NiN_3–Pc–CNT is coordinated by three adjacent N atoms, compared with Ni-N_4 in NiPc–CNT (Supplementary Figures 17e–f and Supplementary Table 8).



Supplementary Figure 17 | (a) XANES-linear fittings of NiN_3–Pc–CNT, NiPc–CNT and reference samples (Ni foil and NiPc). Ni K-edge wavelet transformed extended X-ray absorption fine structure (WT-EXAFS) contour plots of (b) NiN_3–Pc–CNT, (c) NiPc–CNT, and (d) Ni foil. EXAFS fitting in R space and (inset) simulated structure for (e) NiPc–CNT and (f) NiPc with one N-atom vacancy (Ni-N_3-Pc) in NiN_3–Pc–CNT.

Supplementary Table 8 | Ni K-edge EXAFS curve fitting parameters for NiN_3–Pc–CNT and NiPc–CNT.

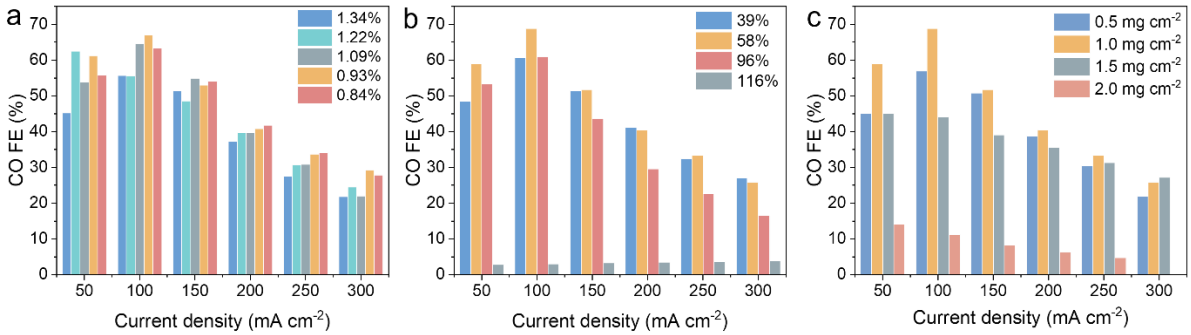
Sample	Condition	Path	N	R (Å)	σ^2 (Å^2)	R factor
NiN_3–Pc–CNT	Air	Ni-N	2.8 (3)	1.84 (2)	0.003 (2)	0.003
NiPc–CNT	Air	Ni-N	3.9 (1)	1.84 (4)	0.003 (2)	0.009

N, coordination number; R, distance between absorber and backscatter atoms; σ^2, Debye- Waller factor to account for both thermal and structural disorder.

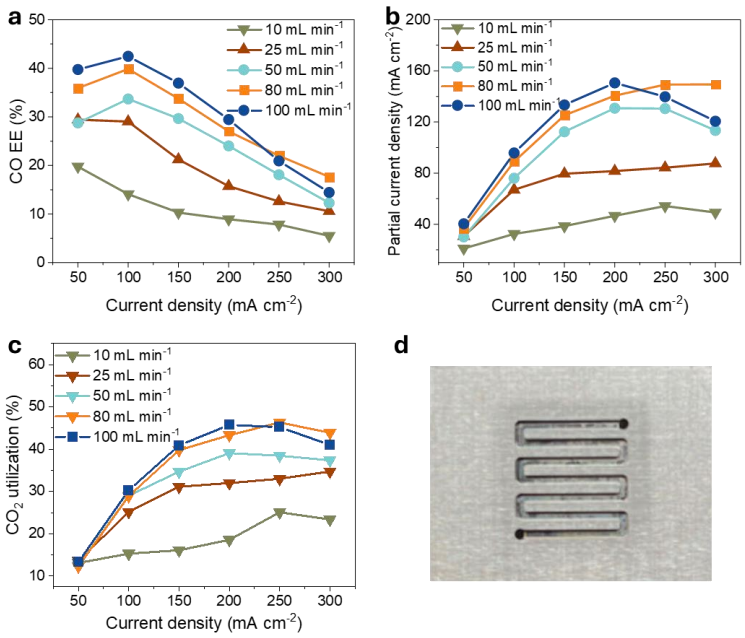
Supplementary Note 6 | Catalyst and system optimizations.

Excess Ni loading in the catalyst may agglomerate to form nanoparticles which will promote hydrogen evolution, but insufficient Ni active sites would result in mass transfer limitations especially at high current densities. Therefore, we prepared NiN_3–Pc–CNT catalysts with different Ni loadings vs. carbon support (Supplementary Figure 18a). We found that the NiN_3–Pc–CNT containing 0.93 wt% nickel has the best performance (~68% faradaic efficiency (FE) at 100 mA cm^{-2}). Additionally, the amount of ionomer in the catalyst layer is important to electrochemical performance since the ionomer functions both as a catalyst binder and ion transfer medium. We then focused on the optimization of ionomer to catalyst weight ratio (Supplementary Figure 18b). The FE starts to rise when the ratio increases from 39% but declines when the ratio is over 58%, illustrating the optimal weight ratio of ionomer to catalyst weight is 58%. It can be explained that less Nafion binder may have limited the ion transfer during the reaction, but excess Nafion may cover the active sites of the catalyst to get low FE. After obtaining the optimal nickel loading and the weight ratio of ionomer to catalyst, we sought to prepare the catalyst layer with different catalyst loadings on the electrode from 0.5 to 2.0 mg cm^{-2} (Supplementary Figure 18c). When the catalyst loading is only 0.5 mg cm^{-2}, small amounts of NiN_3–Pc–CNT induced insufficient active sites for the reaction, but too many catalysts might induce thick catalyst layer and bring mass transfer resistance to get deteriorated performance. The optimal catalyst loading was found at 1.0 mg cm^{-2}. The optimization steps above brought our FE of NiN_3–Pc–CNT to ~70%.

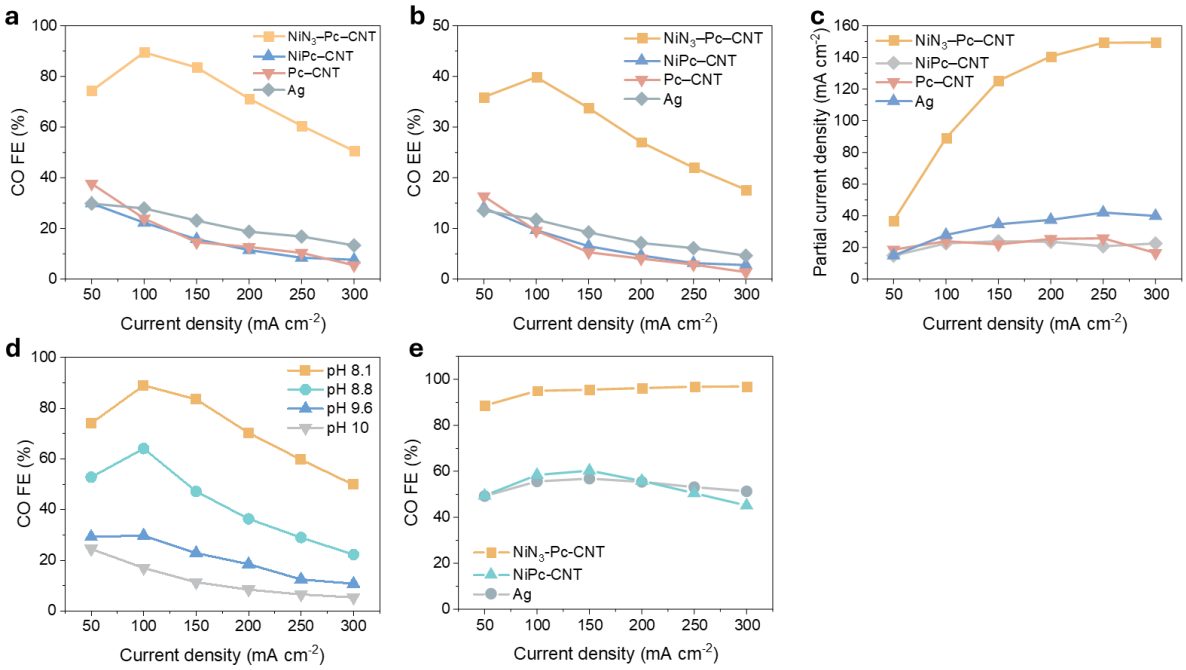
We further investigated the effect of catholyte flow rate on the performance (Figure 4a and Supplementary Figure 19), finding the maximum CO FE of 94% at 100 mL min^{-1} and 90% at 80 mL min^{-1}. However, it seems that higher flowrate gave us higher CO FE, but the CO FE started to decrease rapidly at higher current densities. We believe this performance deterioration is due to the physical detachment of the nickel catalyst when the flowrate is very high (Supplementary Figure 19d), suggesting that mechanical detachment of the catalyst could be one possible failure mechanism.



Supplementary Figure 18 | (a) Optimization of nickel loading in the catalyst. The amount of Nickel loading in NiN_3–Pc–CNT is detected by inductively coupled plasma atomic emission spectroscopy. The catalyst loading on the electrode is 1 mg cm^{-2}. The catholyte flowrate is at 25 ml min^{-1}. The weight ratio of Nafion to catalyst is 58%. (b) Optimization of ionomer loading on the cathode catalyst layer. We prepared the electrodes with different weight ratio of Nafion to catalyst where we found the optimized ratio is 58%. The nickel amount in NiN_3–Pc–CNT is 0.93 wt%. The catholyte flowrate is at 25 ml min^{-1}. The catalyst loading on the electrode is 1 mg cm^{-2}. (c) Optimization of catalyst loading on the electrode. We sprayed different amount of catalyst onto the electrode. The nickel amount in NiN_3–Pc–CNT is 0.93 wt%. The catholyte flowrate is at 25 ml min^{-1}. The weight ratio of Nafion to catalyst is 58%.



Supplementary Figure 19 | Optimization of catholyte flowrate and CO_2 utilization. (a) EE(CO), (b) partial current density and (c) CO_2 utilization using NiN_3–Pc–CNT at different catholyte flowrate in reactive capture. (d) Picture of flow field after 100 mL min^{-1} run (the physical detachment of the catalyst was observed, indicating 100 mL min^{-1} is too high to be used in the reactive capture which might induce the performance deteriorations at high current densities).



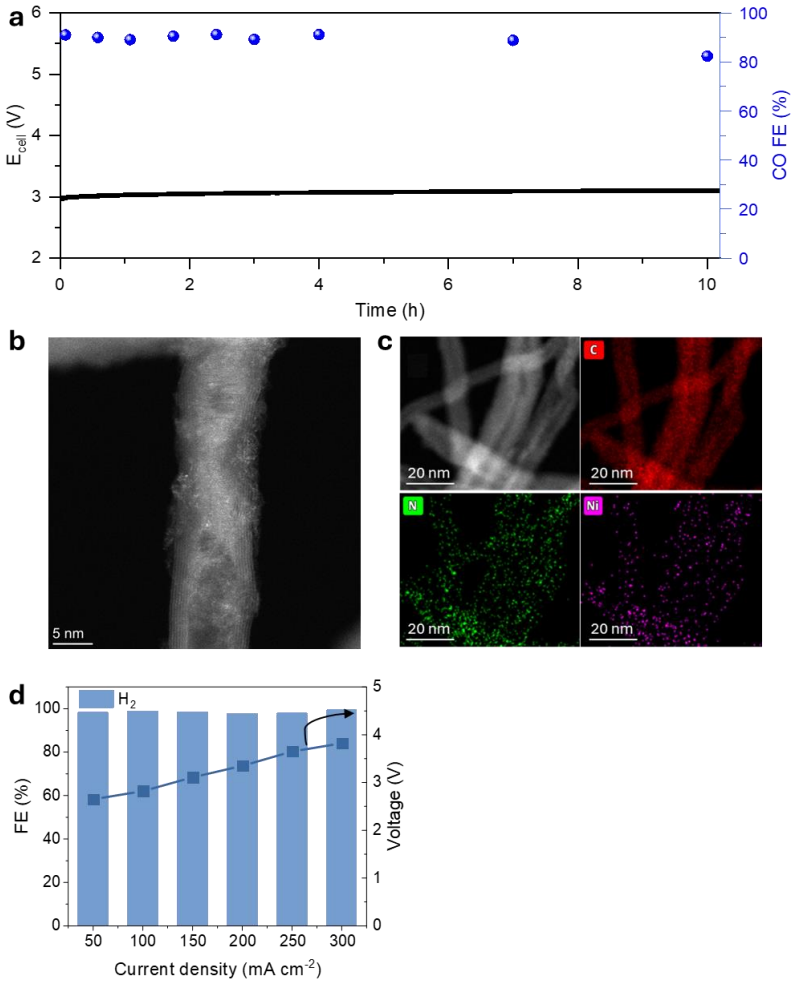
Supplementary Figure 20 | (a) FE, (b) EE and (c) CO partial current density of reactive capture using different catalysts at 80 mL min^{-1}. (d) FE performance for 1 M K-Gly at different pH values. (e) FE of reactive capture using 3 M $KHCO_3$ as the capture solution. As an additional control to examine the active sites in NiN_3–Pc–CNT, we measured the CNT supported phthalocyanine without nickel atoms (Pc–CNT, prepared without addition of nickel source, details in the Methods) and observed a CO FE of 40% (Supplementary Figure 20a), indicating that the Ni centres in NiN_3–Pc–CNT are the primary catalytically active sites, and the nitrogen dopants of Pc is also responsible to catalyse the captured CO_2 to CO.

Supplementary Table 9 | Comparison of NiN_3–Pc–CNT and previously reported catalysts toward the conversion to CO in reactive capture using KOH or amine as the capture solution.

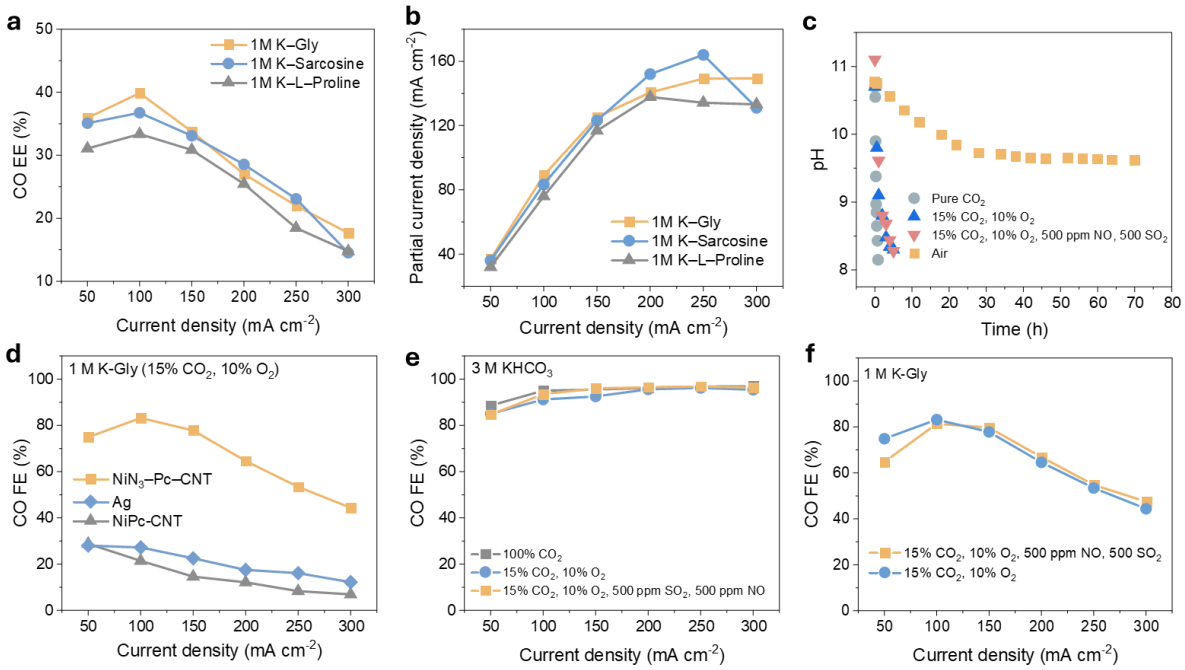
Catalyst	**Capture solution**	**Voltage (V)**	**Current density (mA cm^{-2})**	**Stability (h)**	**FE (%)**	**EE (%)**	**References**
Ag	2 M KOH	3.8	200	23	46	16.1	16
Ag	1 M KOH	3.1	100	145	28	12.0	17
Cu	EDA	–0.78V vs RHE	18.4	-	58	-	18
Ag	30% MEA	–0.8V vs RHE	50	2	72	-	19
Ni-N/C	2 M K-Gly	3.0	50	32	64	31	4
Ni-N/C	5 M MEA	-	50	-	64.9	-	20
coral-Ag/C	3 M TREA	-	20	70	30	13	21
NiN_3–Pc–CNT	1 M K-Gly	2.96	100	12	94	42	This work

Supplementary Table 10 | Comparison of NiN_3–Pc–CNT and other reported catalysts in bicarbonate-based reactive capture systems to produce CO.

Catalyst	**Capture solution**	**Voltage (V)**	**Current density (mA cm^{-2})**	**FE (%)**	**EE (%)**	**References**
Ag	3 M $KHCO_3$	3.6	100	59	21.8	22
Ni-N/C	2 M $KHCO_3$	3.7	150	88	31.6	4
amino-CoPc/CNT	3 M $KHCO_3$	2.75	100	70	33.9	23
CoPc@CNT	3 M $KHCO_3$	3.4	100	96.2	37.6	24
Ni-SAC	3 M $KHCO_3$	3.7	200	93	33.4	25
NiN_3–Pc–CNT	3 M $KHCO_3$	2.83	100	95	44.6	This work
		3.12	200	96.2	41.0	
		3.4	300	96.9	37.9	



Supplementary Figure 21 | (a) Stability test of K-Gly-based reactive capture at 100 mA cm^{-2} and 80 mL min^{-1} catholyte flowrate. (b) HAADF-STEM image and (c) EDS maps (C, red; N, green; Ni, pink) of NiN_3–Pc–CNT after the stability test. (d) FE using NiN_3–Pc–CNT with fresh 1 M K-Gly without CO_2 capture.



Supplementary Figure 22 | (a) EE comparison and (b) partial current density using different post-capture amino acid salt solutions. (c) pH values in capture solutions using different carbon sources—pure CO_2, simulated flue gas (15% CO_2, 10% O_2, N_2 balanced), simulated flue gas containing common contaminates (15% CO_2, 10% O_2, 500 ppm SO_2, 500 ppm NO, N_2 balanced), and air. (d) Performance comparison of Ag, NiPc–CNT and NiN_3–Pc–CNT using simulated flue gas captured 1 M K-Gly solution as the capture solution. (e) FE(CO)s of NiN_3–Pc–CNT using 3 M $KHCO_3$ bubbled with pure CO_2, simulated flue gas or simulated flue gas containing common contaminates as the capture solution. (f) FE(CO)s of NiN_3–Pc–CNT using 1 M post-capture K-Gly from simulated flue gas or simulated flue gas containing common contaminates. The comparable CO FEs of each catalyst using CO_2 (Figure 4b), simulated flue gas (Supplementary Figure 22d–e) or simulated flue gas containing common contaminates (Supplementary Figure 22f) captured K-glycine solutions demonstrated the good flue gas tolerance of NiN_3–Pc–CNT.

Supplementary Note 7 | Energy analysis for the reactive capture system.

This section provides a brief description of the energy assessment model, and the assumptions used to calculate the energy cost to generate one ton of CO, comparing different systems, including gaseous CO_2R, hydroxide-based reactive capture and amino-acid-based reactive capture used in this work. The total energy required to generate one ton of CO is the sum of the energy for CO_2 capture, CO_2 electrolysis, and CO_2 recovery.

CO_2 capture

For the CO_2 capture processes, the model considers the energy required to capture CO_2 (400 ppm) from the air[26]. This process requires 0.22 GJ/ton$_{CO2}$ (61 kWh/ton$_{CO2}$) to operate the contactor fans and 0.08 GJ/ton$_{CO2}$ (21 kWh/ton$_{CO2}$) for fluid pumping[26]. For gaseous CO_2R, a calcium caustic loop is required to generate gaseous CO_2 from the capture solution, requiring a total energy of 6.57 GJ/ton$_{CO2}$ (5.25 GJ/ton$_{CO2}$ + 366 kWh/ton$_{CO2}$). Reactive capture processes bypass the CO_2 regeneration step, thereby eliminating this energy requirement.

For all systems, we consider stoichiometrically that 1.57 tons of CO_2 are required to generate one ton of CO. Therefore, the CO_2 capture costs are calculated as follows:

Gaseous CO_2RR:

$$CO_2\ capture\ cost\ for\ gaseous\ CO_2R\ \left[\frac{GJ}{ton_{CO}}\right]$$
$$= Capture\ cost\ \left[\frac{GJ}{ton_{CO_2}}\right] \times \frac{molecular\ weight_{CO_2}\left[\frac{g}{mol}\right]}{molecular\ weight_{CO}\left[\frac{g}{mol}\right]} \times molar\ ratio$$
$$= 6.57\frac{GJ}{ton_{CO_2}} \times \frac{44\frac{g}{mol}}{28\frac{g}{mol}} \times 1 = 10.32\ \frac{GJ}{ton_{CO}}$$

Reactive capture:

$$CO_2\ capture\ cost\ for\ reactive\ capture\ \left[\frac{GJ}{ton_{CO}}\right]$$
$$= Capture\ cost\ \left[\frac{GJ}{ton_{CO_2}}\right] \times \frac{molecular\ weight_{CO_2}\left[\frac{g}{mol}\right]}{molecular\ weight_{CO}\left[\frac{g}{mol}\right]} \times molar\ ratio$$
$$= 0.3\frac{GJ}{ton_{CO_2}} \times \frac{44\frac{g}{mol}}{28\frac{g}{mol}} \times 1 = 0.47\ \frac{GJ}{ton_{CO}}$$

CO_2 electrolysis

The energy required for CO_2 electrolysis is calculated based on the performance achieved by the CO_2 electrolysis processes. For gaseous CO_2RR, we considered a prior CO_2RR system using acidic electrolytes, which minimizes CO_2 losses and results in the lowest combined energy requirement for electrolysis and CO_2 regeneration. This system achieved a CO FE of 60% at a single-pass conversion of 50% and at a full-cell voltage of 4.0 V[27].

Gaseous CO_2R:

$$CO_2\ electrolysis\ energy\left[\frac{GJ}{ton_{CO}}\right] = total\ charge\ required\left[\frac{C}{ton_{CO}}\right] \times full-cell\ voltage\ [V] \times 10^{-9}\left[\frac{GJ}{J}\right]$$

$$= \frac{\frac{10^6 g}{28\frac{g}{mol}} \times 2 \times 96485\frac{sA}{mol}}{60\ \%} \times 4\,V \times 10^{-9}\frac{GJ}{J} = 45.95\frac{GJ}{ton_{CO}}$$

For the amino-acid-based reactive capture used in this work, we achieved a CO FE of 90% at a CO_2 utilization of 29% and at a full-cell voltage of 2.97 V.

Reactive capture:

$$CO_2\ electrolysis\ energy\left[\frac{GJ}{ton_{CO}}\right] = total\ charge\ required\left[\frac{C}{ton_{CO}}\right] \times full-cell\ voltage\ [V] \times 10^{-9}\left[\frac{GJ}{J}\right]$$

$$= \frac{\frac{10^6 g}{28\frac{g}{mol}} \times 2 \times 96485\frac{sA}{mol}}{94\ \%} \times 2.96\,V \times 10^{-9}\frac{GJ}{J} = 21.94\frac{GJ}{ton_{CO}}$$

CO_2 recovery

In all processes, the unconverted CO_2 is mixed with the product stream and requires recovery before being fed back to the electrolyzer. Conventional amine adsorption using monoethanolamine process is used to capture CO_2 from a concentrated stream. This process consumes 4.0 GJ/ton_{CO2} for heat input to the reboiler, 0.5 GJ/ton_{CO2} for the blower and 0.1 GJ/ton_{CO2} for pumps[28]. For the reactive capture processes, a reboiler is not required as the capture solution can be fed back directly back to the electrolyzer for conversion.

Gaseous CO_2RR:

$$CO_2\ recovery\ cost\ for\ gaseous\ CO_2R\left[\frac{GJ}{ton_{CO}}\right] = Capture\ cost\left[\frac{GJ}{ton_{CO_2}}\right] \times CO_2\ not\ utilized\left[\frac{ton_{CO_2}}{ton_{CO}}\right]$$

$$= 4.6\frac{GJ}{ton_{CO_2}} \times \left(\frac{1.57}{0.5} - 1.57\right)\frac{ton_{CO_2}}{ton_{CO}} = 7.22\ \frac{GJ}{ton_{CO}}$$

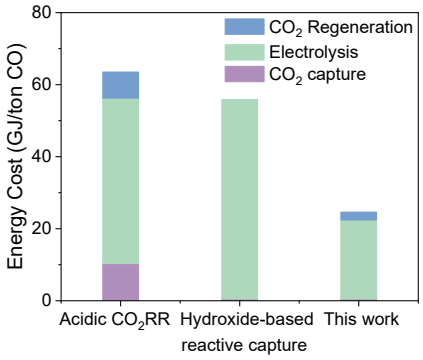
Reactive capture:

$$CO_2\ recovery\ cost\ for\ reactive\ capture\ \left[\frac{GJ}{ton_{CO}}\right] = Capture\ cost\ \left[\frac{GJ}{ton_{CO_2}}\right] \times CO_2\ not\ utilized\ \left[\frac{ton_{CO_2}}{ton_{CO}}\right]$$

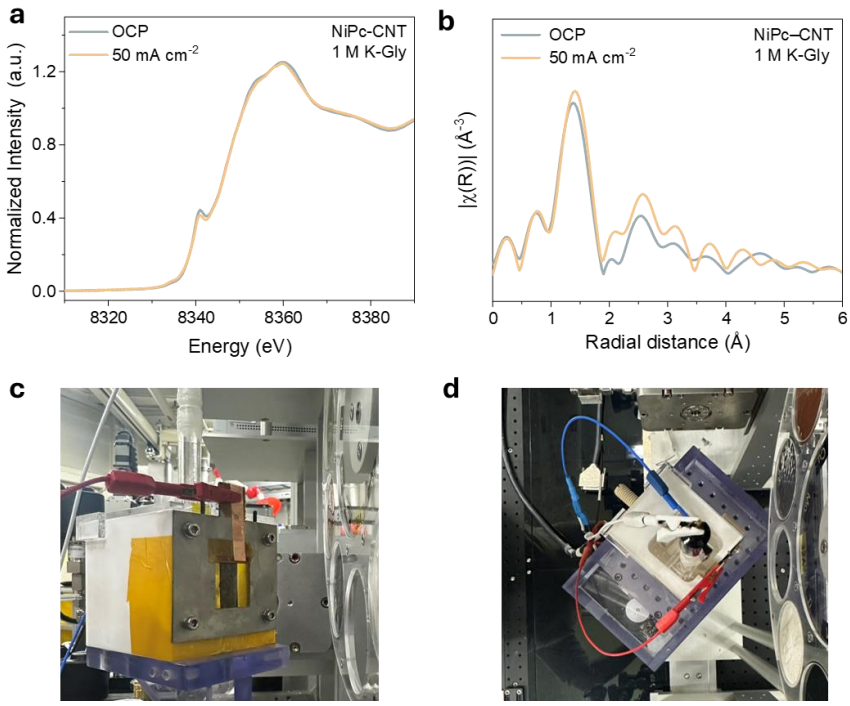
$$= 0.6\frac{GJ}{ton_{CO_2}} \times (\frac{1.57}{0.30} - 1.57)\frac{ton_{CO_2}}{ton_{CO}} = 2.20\ \frac{GJ}{ton_{CO}}$$

Supplementary Table 11 | Energy assessment of producing CO using acidic CO_2RR, hydroxide-based reactive capture, and K^+-glycine based reactive capture system.

Parameters	**Acidic CO_2RR[27]**	**Hydroxide-based reactive capture[16]**	**K^+-glycine reactive capture (this work)**
Cell voltage (V)	4.0	3.7	2.96
CO Faradaic efficiency (%)	60	46	94
CO_2 utilization (%)	50	100	30
CO_2 capture (GJ ton_{CO}^{-1})	10.32	0.47	0.47
CO_2 electrolysis (GJ ton_{CO}^{-1})	45.95	55.43	21.94
CO_2 recovery (GJ ton_{CO}^{-1})	7.23	0	2.20
Overall energy (GJ ton_{CO}^{-1})	**63.50**	**55.91**	**24.61**



Supplementary Figure 23 | Comparison of energy cost towards CO among acidic CO_2 electrolysis, hydroxide- and amino-acid-based reactive capture using techno-economic analysis.



Supplementary Figure 24 | (a) In situ Ni K-edge XANES spectra and (b) the corresponding k^2-weighted Fourier transforms of NiPc–CNT at 50 mA cm^{-2} using 1 M K-Gly as the electrolyzer. (c) Side view and (d) top view images of H-cell photo used for the in situ XAS tests.

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